

HOSPITAL MANAGEMENT SYSTEM

1. Introduction

The **Hospital Management System (HMS)** is a software application developed to manage hospital operations such as patient registration, doctor management, and appointment scheduling. The system helps hospitals maintain records digitally and improves efficiency by automating manual processes.

This project is implemented using **Java**, with a structured folder system including database files and user interface components. The application provides a centralized platform for managing hospital data and streamlining administrative tasks.

Hospital Management Systems are widely used to:

- Store patient information
- Manage doctor details
- Schedule appointments
- Maintain hospital records
- Improve service efficiency

2. Problem Statement

Traditional hospital systems rely heavily on manual record-keeping, which leads to:

- Data loss
- Time-consuming processes
- Human errors
- Difficulty in managing patient records
- Inefficient appointment scheduling

Therefore, a computerized system is required to:

- Automate hospital operations
- Maintain accurate records
- Improve efficiency
- Provide quick access to patient and doctor information

3. Objectives

The main objectives of the Hospital Management System are:

- To manage patient records efficiently
- To maintain doctor information
- To schedule appointments automatically

- To store hospital data securely
- To reduce paperwork
- To improve hospital management

4. Scope of the Project

The system can be used in:

- Hospitals
- Clinics
- Medical centers
- Healthcare institutions

Future enhancements may include:

- Billing system
- Pharmacy management
- Laboratory management
- Online appointment booking
- Report generation

5. System Overview

The Hospital Management System consists of the following main modules:

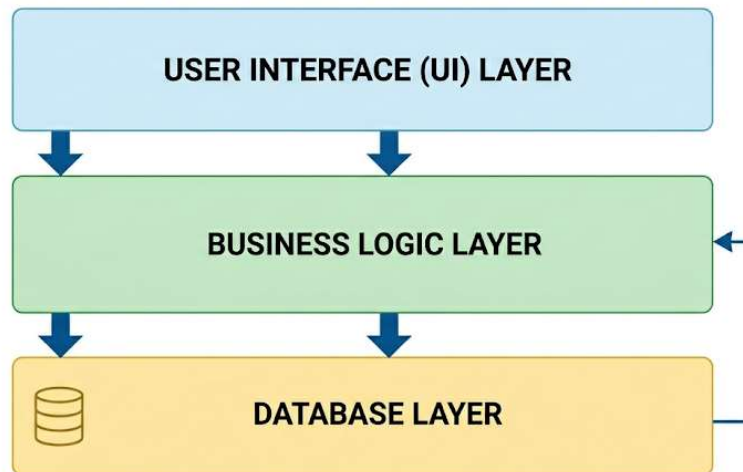
- Login System
- Dashboard
- Patient Management
- Doctor Management
- Appointment Scheduling
- Database Management

The application interface includes screens such as:

- Login screen
- Dashboard
- Patient management screen
- Doctor management screen
- Appointment scheduling screen
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6. System Architecture

SYSTEM ARCHITECTURE: LAYERED ARCHITECTURE TYPE



7. Technologies Used

Technology	Purpose
Java	Application development
JDBC	Database connectivity
MySQL / Database	Data storage
Swing / GUI	User interface
IDE (IntelliJ / Eclipse / NetBeans)	Development environment

8. Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB or more
- Hard Disk: 500 GB
- Monitor: 15-inch display

9. Software Requirements

- Operating System: Windows / Linux / Mac
- Java JDK: Version 8 or higher
- Database: MySQL
- IDE: Eclipse / IntelliJ / NetBeans

10. Modules Description

10.1 Login Module

Purpose:

- Authenticate users
- Provide secure system access

Functions:

- Username validation
- Password verification
- Login authentication

10.2 Patient Management Module

Purpose:

- Store patient information

Functions:

- Add patient
- Update patient details
- Delete patient
- View patient records

10.3 Doctor Management Module

Purpose:

- Manage doctor information

Functions:

- Add doctor
- Update doctor details
- Delete doctor
- View doctor records

10.4 Appointment Scheduling Module

Purpose:

- Schedule appointments between patients and doctors

Functions:

- Book appointment
- Update appointment
- Cancel appointment
- View appointment list

10.5 Dashboard Module

Purpose:

- Provide overview of system data

Functions:

- Display system statistics
- Quick navigation
- System summary
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11. Database Design

Tables Used

- **Patient Table**

Field	Type
Patient_ID	
Name	
Age	
Gender	
Contact	
Address	

- **Doctor Table**

Field	Type
Doctor_ID	
Name	
Specialization	
Contact	

- **Appointment Table**

Field	Type
Appointment_ID	
Patient_ID	
Doctor_ID	
Date	
Time	

12. UML Diagrams

Use Case Diagram

Actors:

- Admin
- Doctor
- Patient

Use Cases:

- Login
- Manage Patients
- Manage Doctors
- Schedule Appointment
- View Dashboard

Class Diagram

Classes:

- Patient
- Doctor
- Appointment
- User
- Database

13. Advantages

- Easy data management
- Fast record retrieval
- Reduced paperwork
- Improved efficiency
- Secure data storage

15. Future Enhancements

- Online appointment booking
- Billing system
- Pharmacy management
- Laboratory system
- Report generation
- Mobile application

16. Conclusion

The Hospital Management System successfully automates hospital operations such as patient management, doctor management, and appointment scheduling. The system improves efficiency, reduces manual work, and provides secure data storage.

This project demonstrates practical implementation of Java-based software development and database management in a real-world healthcare environment.

17. Source Code

```
package com.hospital;

import com.hospital.ui.LoginFrame;
import javax.swing.SwingUtilities;
import javax.swing.UIManager;

public class Main {

    public static void main(String[] args) {

        try {

            // Set system look and feel for better UI

            UIManager.setLookAndFeel(UIManager.getSystemLookAndFeelClassName());

        } catch (Exception e) {

            e.printStackTrace();

        }

        // Launch the application on EDT

        SwingUtilities.invokeLater(() -> {

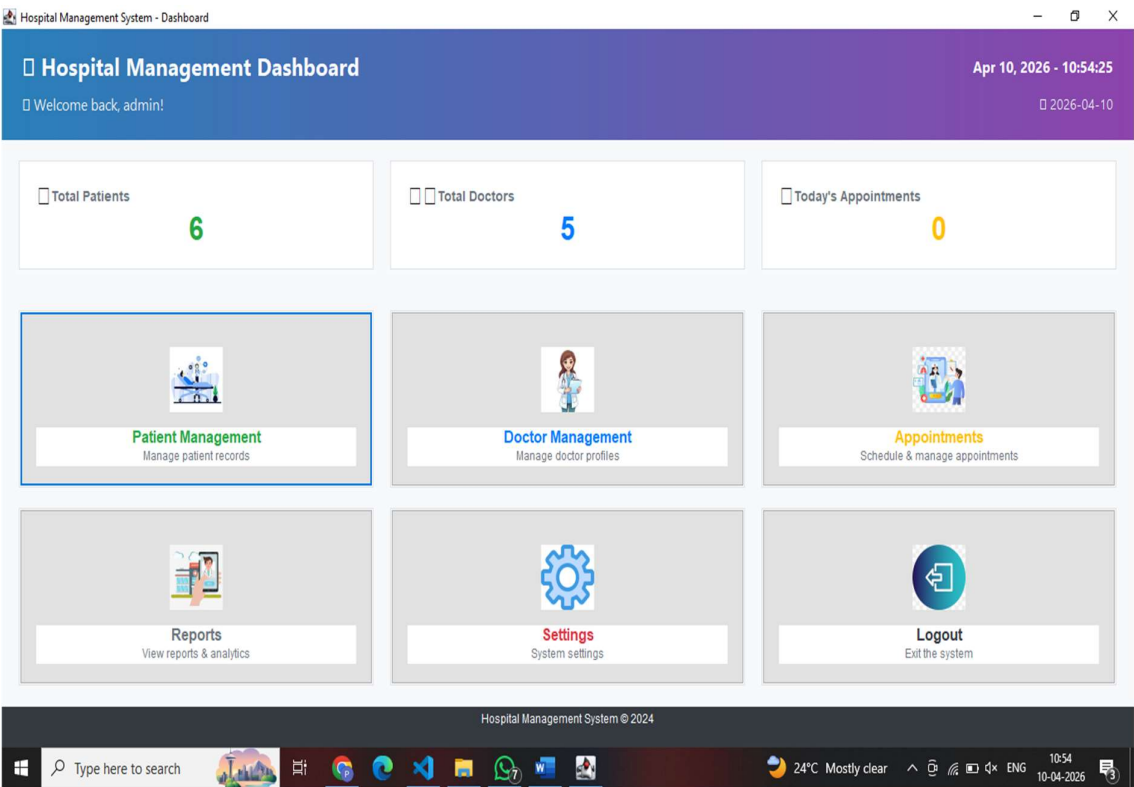
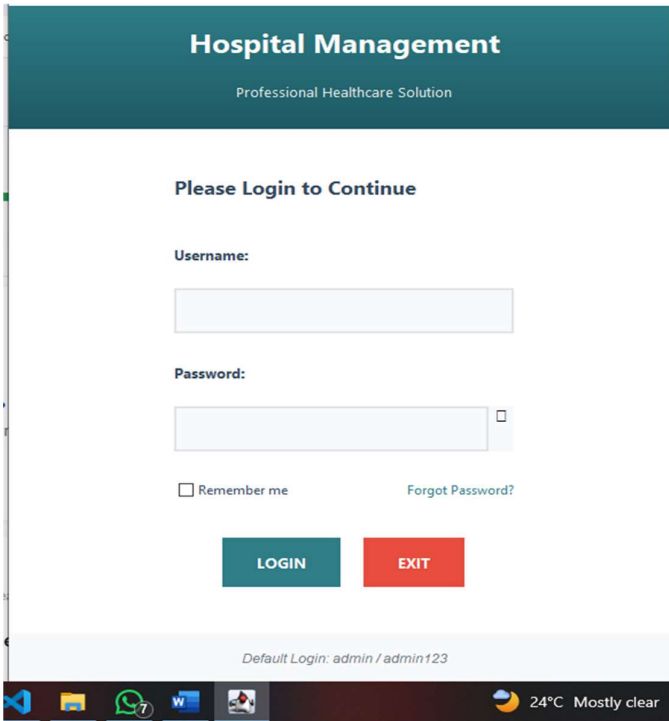
            new LoginFrame().setVisible(true);

        });

    }

}
```

18. Screenshots



Patient Management

Search:

Search

Add Patient

Edit Patient

Delete Patient

Refresh

ID	Name	Age	Gender	Phone	Email	Disease	Blood Group	Admission Date
6	Harshitha	21	Female	889546321	harshitha@gmail.com	blood deficiency	A+	2026-03-24
5	Eva Brown	29	Female	8765432105	eva.brown@email.com	Skin Allergy	A-	2024-02-15
4	David Wilson	55	Male	8765432106	david.w@email.com	Arthritis	AB+	2024-02-10
3	Carol Davis	32	Female	8765432107	carol.d@email.com	Migraine	O+	2024-02-01
2	Bob Smith	45	Male	8765432108	bob.smith@email.com	Diabetes	B+	2024-01-20
1	Alice Johnson	28	Female	8765432109	alice.j@email.com	Hypertension	A+	2024-01-15

Type here to search

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11:02 10-04-2026

Appointment Management

Search:

Search

Status: All

Filter

Book Appointment

Edit Appointment

Cancel Appointment

Delete Appointment

Refresh

Book New Appointment

Patient *: Harshitha (ID: 6)

Doctor *: Dr. Dr. Emily Davis (Dermatology)

Date (YYYY-MM-DD) *: 2026-04-10

Time (HH:MM) *: 10:00

Status: Scheduled

Notes:

Check Availability

Date format: YYYY-MM-DD (e.g., 2024-03-15)

Time format: HH:MM (e.g., 14:30)

Save

Cancel

ID	Patient	Doctor	Status	Notes
----	---------	--------	--------	-------

Doctor Management

Search:

Search

Add Doctor

Edit Doctor

Delete Doctor

Refresh

ID	Name	Specialization	Phone	Email	Experience	Qualification	Fee	Available Days	Available Time
4	Dr. Emily Davis	Dermatology	9876543213	emily.davis@hospital...	10 years	MD Dermatology	\$450.00	Mon,Tue,Thu,Fri	11:00-18:00
1	Dr. John Smith	Cardiology	9876543210	john.smith@hospital...	15 years	MD Cardiology	\$500.00	Mon,Tue,Wed,Thu,Fri	09:00-17:00
3	Dr. Michael Brown	Orthopedics	9876543212	michael.brown@hospit...	18 years	MS Orthopedics	\$600.00	Tue,Thu,Sat	08:00-14:00
5	Dr. Robert Wilson	Neurology	9876543214	robert.wilson@hospit...	20 years	DM Neurology	\$700.00	Wed,Thu,Fri,Sat	09:00-15:00
2	Dr. Sarah Johnson	Pediatrics	9876543211	sarah.johnson@hosp...	12 years	MD Pediatrics	\$400.00	Mon,Wed,Fri,Sat	10:00-16:00

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